# Statistical Evaluation of Cross-Sectional Equity Trading Signals

## 1 Introduction

When backtesting cross-sectional equity signals, it is deceptively easy to generate impressive in-sample Sharpe ratios simply by tweaking lookback windows, transformation styles, or universe filters. The real challenge is determining whether an observed alpha reflects genuine predictive power or is merely an artifact of multiple testing, factor redundancy, and backtest overfitting.

This study evaluates how multiple testing corrections, factor residualization, benchmark factor exposures, and strict out-of-sample locking alter the apparent predictive performance of equity trading signals. Rather than asking which signal yields the highest historical return, we examine whether a disciplined combination of statistically screened, uncorrelated signals offers better out-of-sample stability than blindly picking the top historical winner.

Instead of treating strategy portfolios as the primary end product, we view them as diagnostic tools to test signal validity. Across every stage—raw vs. adjusted p-values, unadjusted vs. factor-residualized returns, single-winner selection vs. independent signal blending, and gross vs. net performance—we apply a consistent benchmark: a strategy isn't successful just because its backtest line goes up and to the right.

## 2 Data and Experimental Design

### 2.1 Reproducible Synthetic Panel

To establish a controlled test environment, we generate a synthetic monthly panel of 240 assets spanning January 2003 through December 2022. Applying lagged investability and liquidity filters leaves a final research sample of 56,022 usable asset-month observations starting in February 2003. The underlying return generator embeds weak momentum, short-term reversal, and low-volatility effects. Crucially, the signal coefficients are intentionally attenuated in the final regime to simulate a harsher, lower-alpha out-of-sample market environment.

The research pipeline splits the timeline into two isolated periods:

- **In-Sample (Training):** February 2003 through December 2016 (167 signal formation months).
- **Out-of-Sample (Locked Test):** January 2017 through November 2022 (71 signal formation months; forward returns realized February 2017 through December 2022).

Setting a fixed seed (`20260824`) ensures the entire pipeline reproduces identically.

### 2.2 CRSP Market Data Pipeline

For production applications, the preferred empirical source is monthly CRSP data via WRDS. The data pipeline filters for common shares (share codes 10 and 11) on NYSE, AMEX, and NASDAQ, maintaining PERMNO as the primary identifier. Total monthly returns combine standard cash returns and delisting adjustments:

$$R_{i,t}^{\text{total}} = (1 + R_{i,t})(1 + R_{i,t}^{\text{delist}}) - 1.$$

Filters for stock price and market capitalization are strictly lagged by one month. To prevent survivorship bias, downloader scripts targeting static current-day ticker lists are deliberately excluded; delisted firms and historical identifiers are preserved throughout.

## 2.3 Leakage Controls

To eliminate lookahead bias, all signal calculations at month $t$ rely strictly on information available on or before month $t$. Rolling features are shifted prior to computation, and all pre-processing steps—including winsorization limits, PCA loadings, factor residualizations, and signal selection thresholds—are fit exclusively on the training sample before being applied out of sample. The locked test set is accessed exactly once for final verification.

# 3 Signal Construction

We construct 54 candidate signals across seven broad signal families by applying three standardization variants to each base feature: cross-sectional $z$-scores, zero-centered percentile ranks, and size-neutral residualized $z$-scores.

**Table 1:** Prespecified signal grid across 7 signal families (54 candidate specifications total).

| Family | Definition | Windows | Economic Rationale |
|---|---|---|---|
| Momentum | Cumulative lagged return, 1-month skip | 3, 6, 9, 12 | Higher past performance |
| Reversal | Negative recent cumulative return | 1, 2 | Overreaction to short-term losses |
| Volatility | Negative realized standard deviation | 6, 12, 24 | Low-volatility anomaly |
| Downside risk | Negative downside standard deviation | 6, 12 | Downside risk premium |
| Size | Negative log market capitalization | 1 | Small-cap premium |
| Turnover | Negative mean log volume | 3, 6, 12 | Liquidity premium |
| Volume shock | Negative absolute volume deviation | 3, 6, 12 | Low unexpected volume |

For a momentum window $L$ with a 1-month skip, the feature is defined as:

$$\mathrm{MOM}_{i,t}^{(L)} = \prod_{j=2}^{L+1} (1 + R_{i,t-j+1}) - 1.$$

Continuous signals are winsorized monthly at the 1st and 99th percentiles. Cross-sectional $z$-scores are computed as:

$$z_{i,t}^{(k)} = \frac{x_{i,t}^{(k)} - \bar{x}_t^{(k)}}{s_t^{(k)}},$$

and size-neutral variants are generated as residual vectors $u_{i,t}^{(k)}$ from monthly cross-sectional regressions against size $z$-scores:

$$z_{i,t}^{(k)} = a_t + b_t z_{i,t}^{\text{size}} + u_{i,t}^{(k)}.$$

Building multiple specifications of related economic concepts is an intentional feature of the experimental design—it forces our testing framework to deal directly with candidate signal redundancy and researcher degrees of freedom.

# 4 Predictive Performance & Portfolio Construction

## 4.1 Information Coefficients

We evaluate baseline predictive skill using the monthly rank Information Coefficient (IC):

$$IC_t^{(k)} = \mathrm{corr}_{\text{Spearman}}\left(x_{i,t}^{(k)}, R_{i,t+1}\right).$$

We track the time-series mean $IC$, Newey-West HAC $t$-statistics, hit rates (percentage of positive months), and forward IC decay over horizons from 1 to 6 months.

### 4.2 Fama-MacBeth Regressions

Monthly cross-sectional regressions evaluate signal slope coefficient persistence:

$$R_{i,t+1} = \alpha_t + \beta_t^{(k)} x_{i,t}^{(k)} + \varepsilon_{i,t+1}.$$

We test whether the time-series mean of $\beta_t^{(k)}$ is significantly different from zero using Newey-West standard errors. When running a joint regression on six correlation-screened candidates:

$$R_{i,t+1} = \alpha_t + \sum_{k \in \mathcal{S}} \beta_t^{(k)} x_{i,t}^{(k)} + \varepsilon_{i,t+1},$$

only the 6-month size-neutral momentum candidate retains individual statistical significance ($t = 3.41$). The remaining signals lose significance once pitted against one another, illustrating how univariate significance often collapses in multivariate settings due to signal collinearity.

### 4.3 Long-Short Decile Portfolios

Each month, stocks are sorted into deciles based on signal ranks. Portfolios buy the top decile and short the bottom decile using equal weights:

$$R_{t+1}^{LS,(k)} = R_{t+1}^{\text{top},(k)} - R_{t+1}^{\text{bottom},(k)}.$$

One-way turnover is measured as:

$$\text{TO}_t^{(k)} = \frac{1}{2} \sum_i \left| w_{i,t}^{(k)} - w_{i,t}^{(k),\text{pre}} \right|,$$

and net returns account for transaction costs assuming 10 bps per unit of turnover:

$$R_t^{\text{net},(k)} = R_t^{\text{gross},(k)} - 0.001 \, \text{TO}_t^{(k)}.$$

## 5 Multiple Testing Corrections

To account for evaluating 54 candidate signals simultaneously, we compare four statistical decision criteria using Newey-West long-run variance estimators (6 lags):

1. **Naive Unadjusted:** Reject $H_0$ if $p_k < 0.05$.
2. **Bonferroni:** Reject $H_0$ if $p_k < 0.05/K$, controlling familywise error rate (FWER).
3. **Benjamini-Hochberg (BH):** Control False Discovery Rate (FDR) by ranking $p_{(1)} \leq \cdots \leq p_{(K)}$ and finding the maximum $j$ such that $p_{(j)} \leq \frac{j}{K}\alpha$.
4. **Block Bootstrap Max $|t|$:** Center strategy returns under $H_0$, resample 6-month blocks across the signal matrix to preserve cross-signal correlation structures, and construct the empirical distribution of $\max_k |t_k|$.

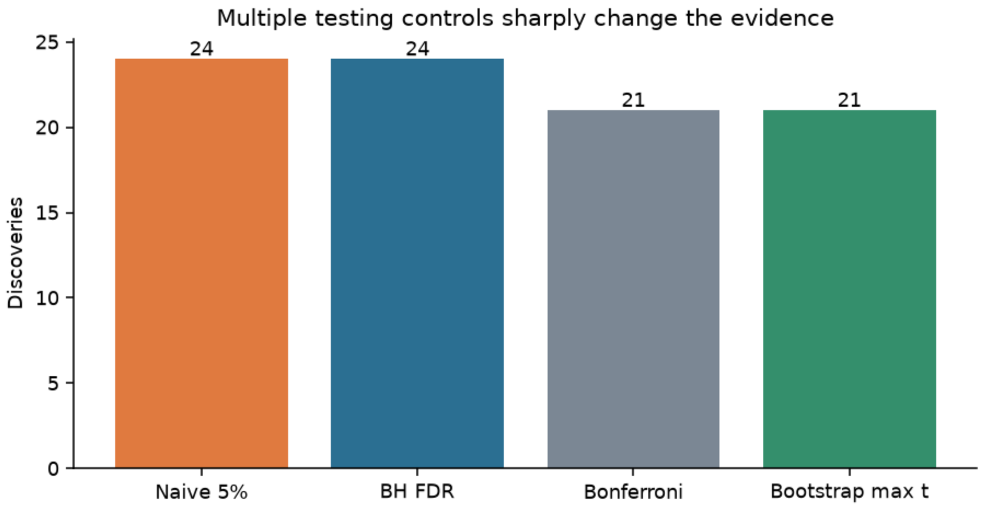


**Figure 1:** In-sample discovery counts across decision rules. Both naive and BH rules retain 24 candidates, whereas Bonferroni and the dependence-aware block bootstrap max-$|t|$ rule filter the set down to 21.

In this setup, BH and naive criteria yield identical survivor counts because the planted signals are strong enough that their $p$-values sit well below the adaptive FDR threshold. FWER controls (Bonferroni and bootstrap) prove more restrictive, pruning three marginal signals.

# 6 Factor Residualization and Redundancy

To inspect underlying redundancy among candidate signals, we perform Principal Component Analysis (PCA) on the standardized signal return matrix $R$:

$$\widehat{\Sigma}_R = V \Lambda V^\top.$$

We run PCA both on raw signal returns and after stripping out standard risk factor exposures (Market, Size, Momentum) via time-series regressions:

$$R_t^{(k)} = a_k + b_{k,M} MKT_t + b_{k,S} SMB_t + b_{k,U} MOM_t + e_t^{(k)}.$$

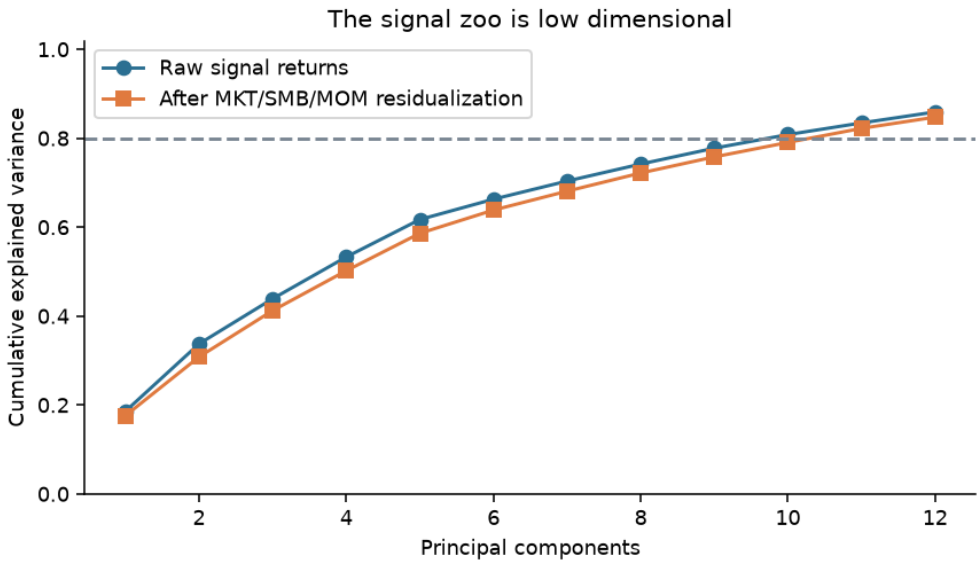


**Figure 2:** Cumulative variance explained by principal components before and after factor residualization. The top component explains 18.6% of raw return variance and 17.6% of residual variance. Reaching 80% variance requires 10 raw components or 11 residual components.

The scree analysis confirms that while candidate signals are highly collinear within feature families, the overall signal grid isn't purely one-dimensional. Stripping standard factor exposures reduces shared variance slightly, but substantial cross-signal correlation remains.

# 7 Out-of-Sample Results

We freeze four selection rules based strictly on training sample performance:

- **In-Sample Winner:** Select the single signal with the highest in-sample HAC $t$-statistic.
- **BH Independent Blend:** Equally weight positive BH discoveries selected greedily to enforce pairwise training correlation $< 0.70$.
- **All In-Sample Positive:** Equally weight all signals with positive in-sample mean returns.
- **All Prespecified Grid:** Equally weight all 54 signals in the candidate grid.

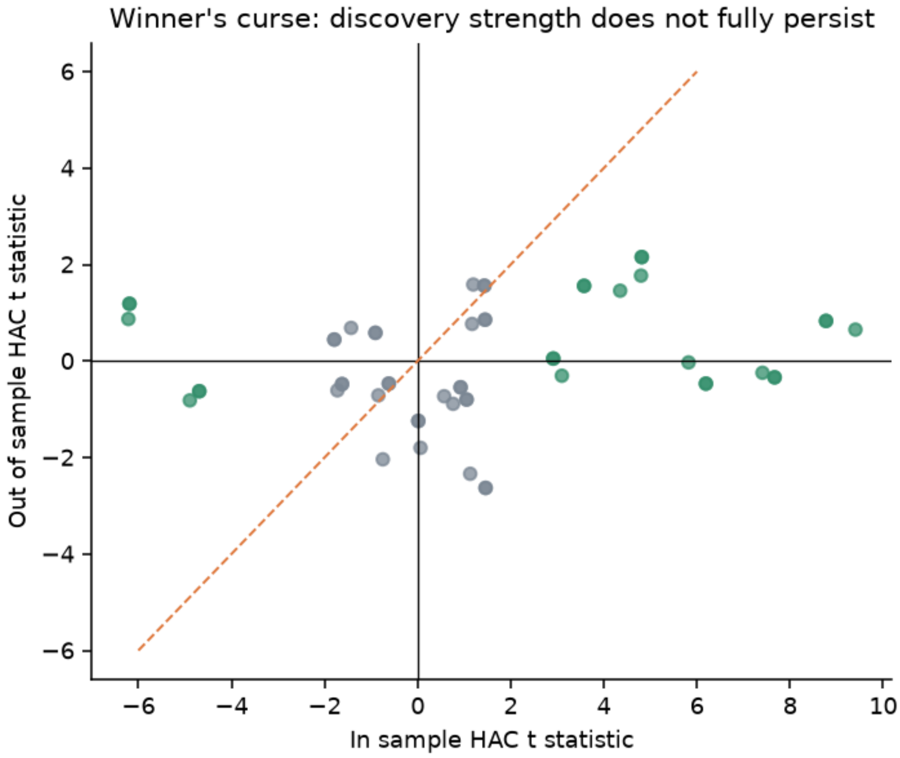


**Figure 3:** In-sample vs. locked out-of-sample HAC $t$-statistics. Green dots mark BH discoveries. Performance drops sharply across nearly all top in-sample performers once evaluated on unseen test data.

**Table 2:** Locked out-of-sample net strategy performance (February 2017 – December 2022).

| Strategy Rule | Ann. Return | Ann. Vol | Sharpe Ratio | Max DD | HAC $t$ |
|---|---|---|---|---|---|
| In-sample winner | 2.10% | 9.78% | 0.26 | −19.32% | 0.65 |
| BH independent blend | 1.96% | 6.68% | 0.32 | −16.84% | 0.68 |
| All positive in-sample | 0.25% | 4.60% | 0.08 | −16.96% | 0.16 |
| All prespecified candidates | 0.05% | 3.28% | 0.03 | −12.21% | 0.06 |

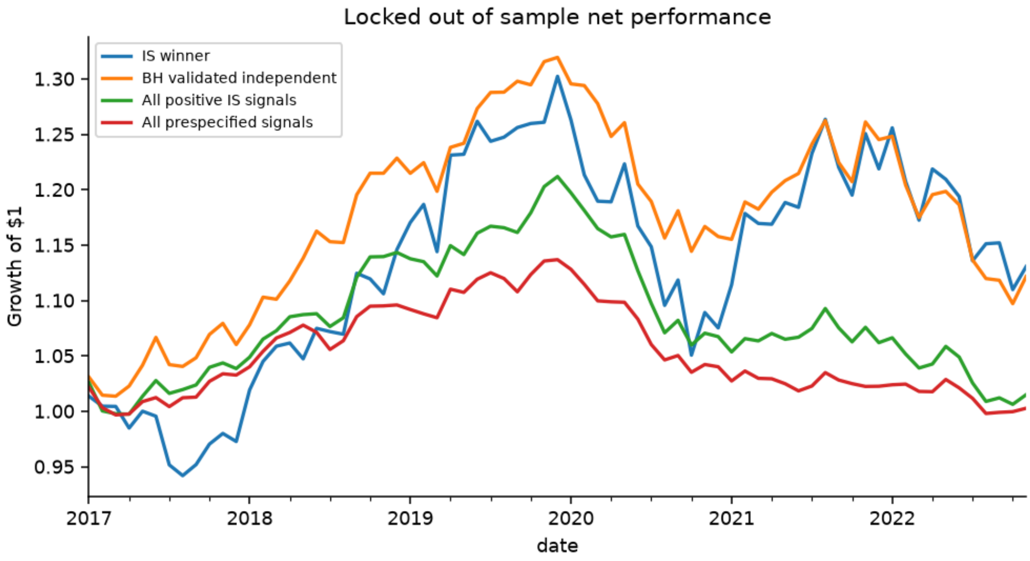


**Figure 4:** Growth of \$1 dollar out of sample after deducting 10 bps linear transaction costs.

The top in-sample signal, `mom_6_1__size_neutral`, sees its $t$-statistic collapse from 9.41 in-sample down to 0.65 out-of-sample (Sharpe ratio 0.26). Combining uncorrelated, validated signals yields a modestly better risk profile—a 0.32 Sharpe ratio, lower volatility (6.68% vs. 9.78%), and shallower drawdown (−16.84% vs. −19.32%)—though neither rule achieves statistical significance out of sample ($t < 1.0$).

# 8 Signal Decay and Turnover Drag

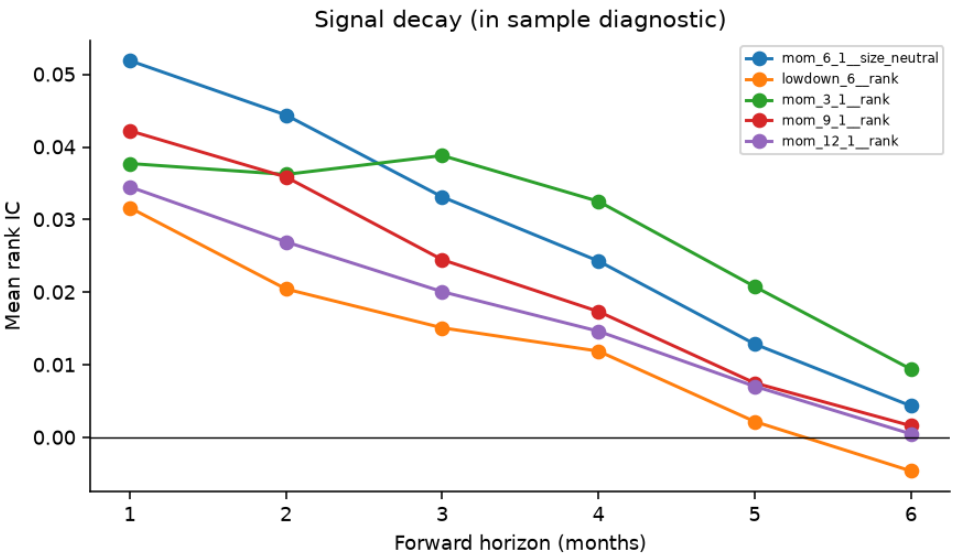


**Figure 5:** In-sample rank IC decay for selected independent signals. Predictive power drops rapidly toward zero by month six.

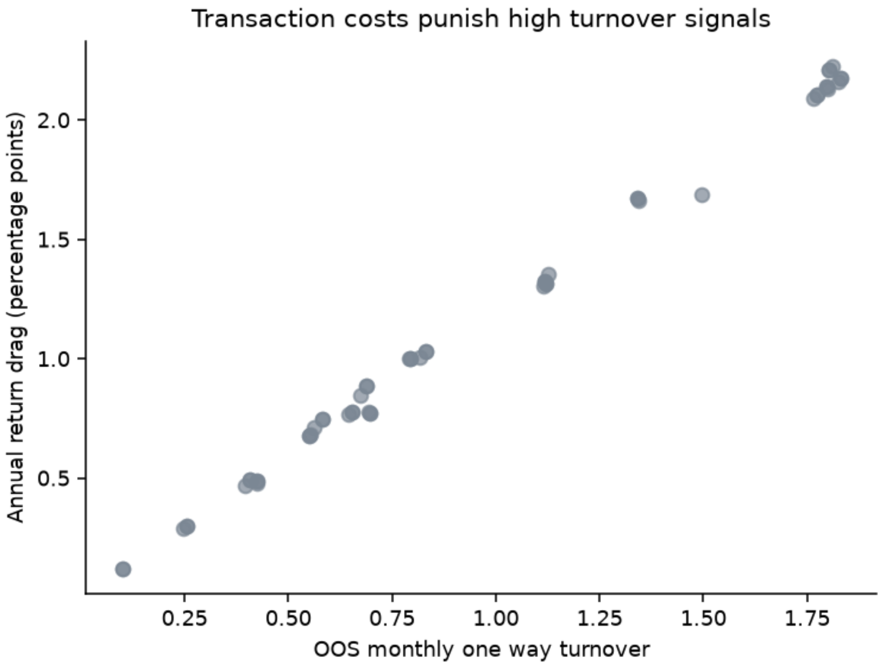


**Figure 6:** Out-of-sample turnover vs. annual performance drag under a fixed 10 bps linear cost model ($r = 0.998$).

Short-horizon features incur heavy turnover costs. High-turnover signals (such as short-term reversal or short-window volume shocks) suffer an annual return drag exceeding 200 bps under 10 bps execution costs. In practice, full implementation models must also factor in short borrow fees, market impact, and capacity constraints.

# 9 Summary of Results

**Table 3:** Key empirical takeaways.

| Topic | Empirical Finding |
|---|---|
| Multiple Testing | Naive and BH retain 24 signals; Bonferroni and max-$\|t\|$ bootstrap retain 21. |
| Winner Decay | Historical winner $t$-statistic drops from 9.41 (IS) to 0.65 (OOS). |
| Factor Structure | 10 raw components explain 80.9% of variance; 11 residual components explain 82.3%. |
| Turnover Drag | Out-of-sample turnover and cost drag correlate at 0.998 under linear costs. |
| Signal Blending | Combining uncorrelated signals boosts OOS Sharpe from 0.26 to 0.32 over single winner. |

# 10 Practical Considerations & Extensions

When transitioning this methodology to live production or real market data, keep the following considerations in mind:

- **Data Integrity:** Standard CRSP data requires explicit handling for delisting returns, point-in-time universe definitions, and microcap exclusions to avoid survivorship bias.
- **Execution Modeling:** Fixed linear transaction cost models are useful baseline proxies, but practical implementation requires incorporating non-linear price impact curves and short-borrow availability.
- **Multiple Testing Choices:** When candidate signals are highly correlated, false discovery rate (FDR) methods provide a practical balance, whereas FWER controls can be overly conservative.

Recommended extensions include testing industry-neutralized signals, market-cap weighting long/short legs, running false discovery Monte Carlo simulations, and validating across the Open Source Asset Pricing (OSAP) empirical dataset.

## 11 Conclusion

Evaluating trading signals requires rigorous statistical guardrails to separate true predictive signal from noise. Standard $p$-values and high historical Sharpe ratios often reflect backtest overfitting rather than structural market inefficiencies.

Applying multiple testing corrections, stripping factor exposures, accounting for transaction costs, and evaluating strategies on locked out-of-sample data provides a realistic view of strategy performance. While combining uncorrelated signals helps smooth out-of-sample returns, no signal selection method can magically rescue strategies where underlying predictive power has faded. A robust quantitative research framework must be built with the capacity to conclude that an apparent strategy advantage lacks statistical support.

## References


[1] Y. Benjamini and Y. Hochberg, "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing," *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.

[2] E. F. Fama and J. D. MacBeth, "Risk, Return, and Equilibrium: Empirical Tests," *Journal of Political Economy*, vol. 81, no. 3, pp. 607–636, 1973.

[3] C. R. Harvey, Y. Liu, and H. Zhu, "... and the Cross Section of Expected Returns," *Review of Financial Studies*, vol. 29, no. 1, pp. 5–68, 2016.

[4] R. D. McLean and J. Pontiff, "Does Academic Research Destroy Stock Return Predictability?" *Journal of Finance*, vol. 71, no. 1, pp. 5–32, 1996.

[5] W. K. Newey and K. D. West, "A Simple, Positive Semidefinite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.

[6] R. Novy Marx and M. Velikov, "A Taxonomy of Anomalies and Their Trading Costs," *Review of Financial Studies*, vol. 29, no. 1, pp. 104–147, 2016.